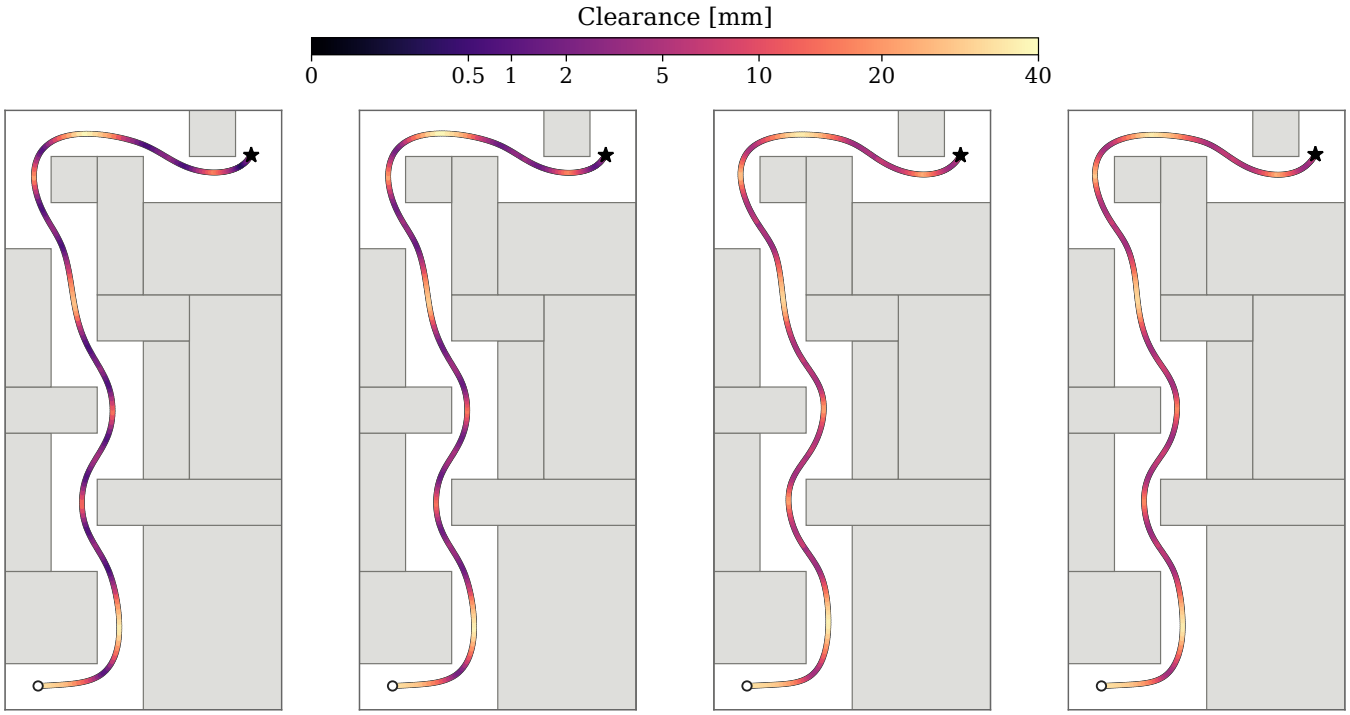
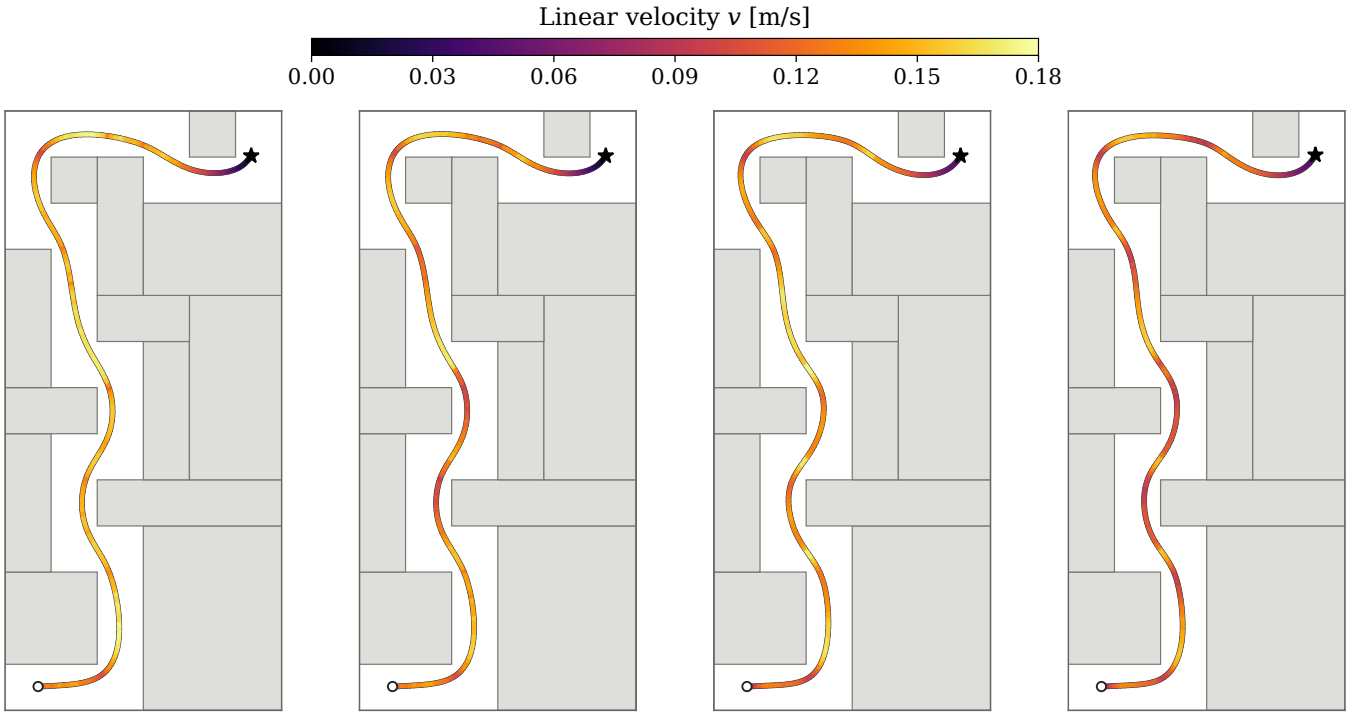


(a) MPCC (b) CC-ECS (c) CC-SF (d) CC-MF

Figure 1: Clearance along the closed-loop trajectories in the maze for a differential-drive robot with a 150×70 mm rectangular footprint. Each method shows one trial with the same initial pose (seed 0). Clearance is the minimum of the exact footprint–obstacle distance and the signed map-boundary margin. All panels share the magma color scale over 0–40 mm. A power-law color normalization ($\gamma = 0.35$) is used to emphasize small clearances. Open circles and filled black stars mark the start and end positions, respectively.



(a) MPCC (b) CC-ECS (c) CC-SF (d) CC-MF

Figure 2: Applied linear velocity along the closed-loop trajectories for the same maze trials (seed 0, differential-drive robot, 150×70 mm rectangular footprint). Each trajectory segment is colored by the linear velocity applied during that interval. All panels share a linear color scale over 0–0.18 m/s. Open circles and filled black stars mark the start and end positions, respectively.